

Proton range calibration for the R³B-CALIFA calorimeter

Mrunmoy Jena

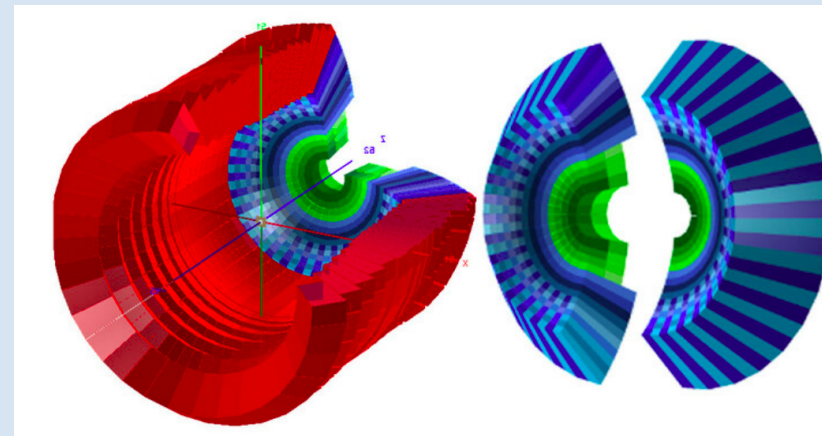
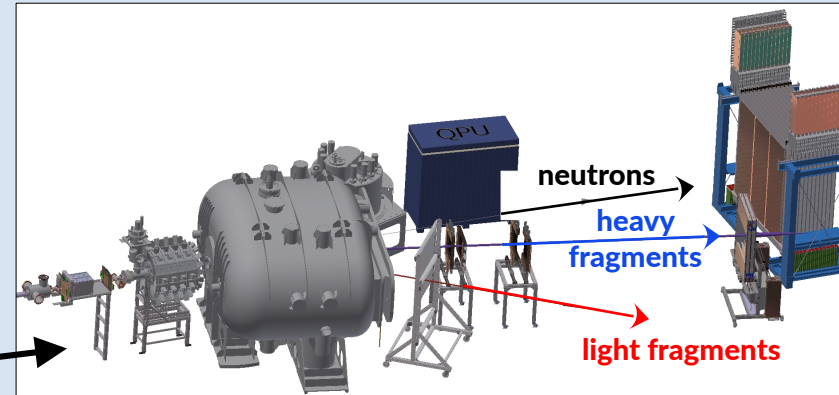
DPG Spring Meeting, Köln
HK 55.2
13.3.2025

Overview of R³B and CALIFA
Structure of calibration algorithm
Calibration in the gamma range
Extrapolation to proton range
QC plots

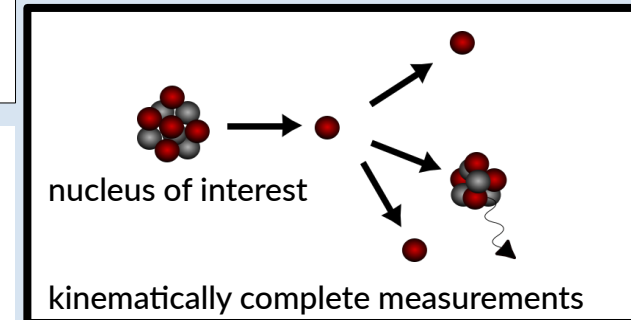
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TUM Members:
Mrunmoy Jena, Tobias Jenegger, Roman Gernhäuser

R³B: Reactions with Radioactive Beams



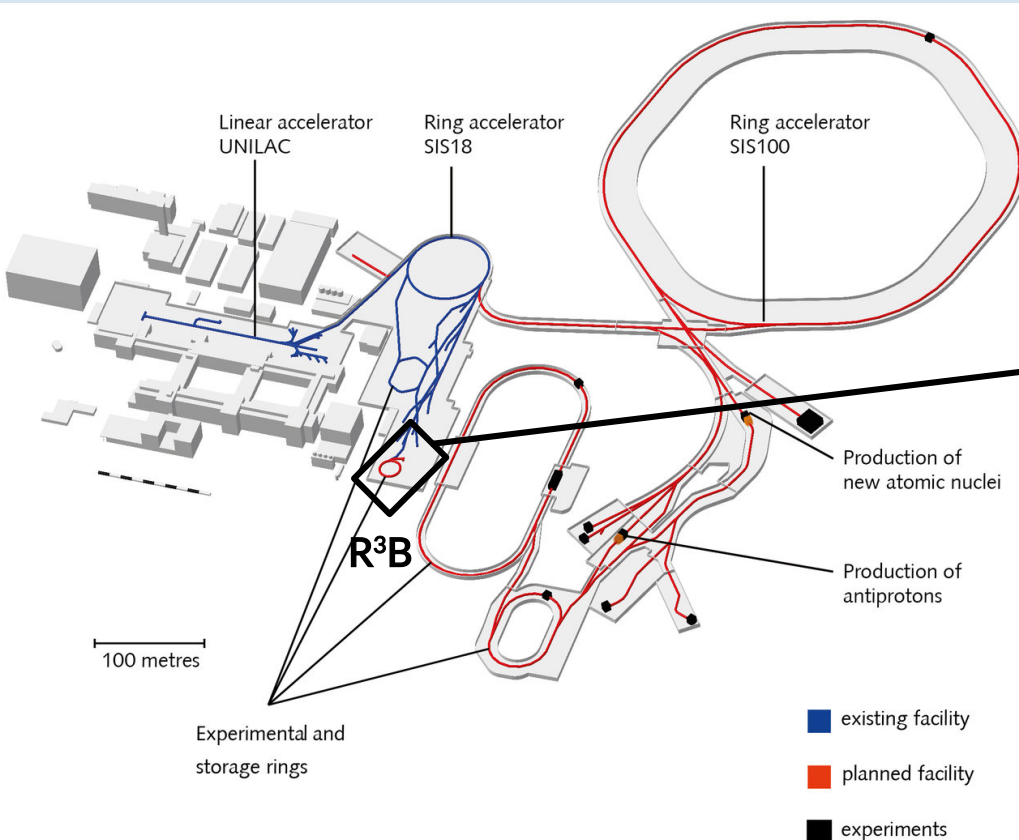
CALorimeter for the In Flight detection
of γ rays and light charged pArticles



Barrel: $43^\circ < \theta < 140^\circ$

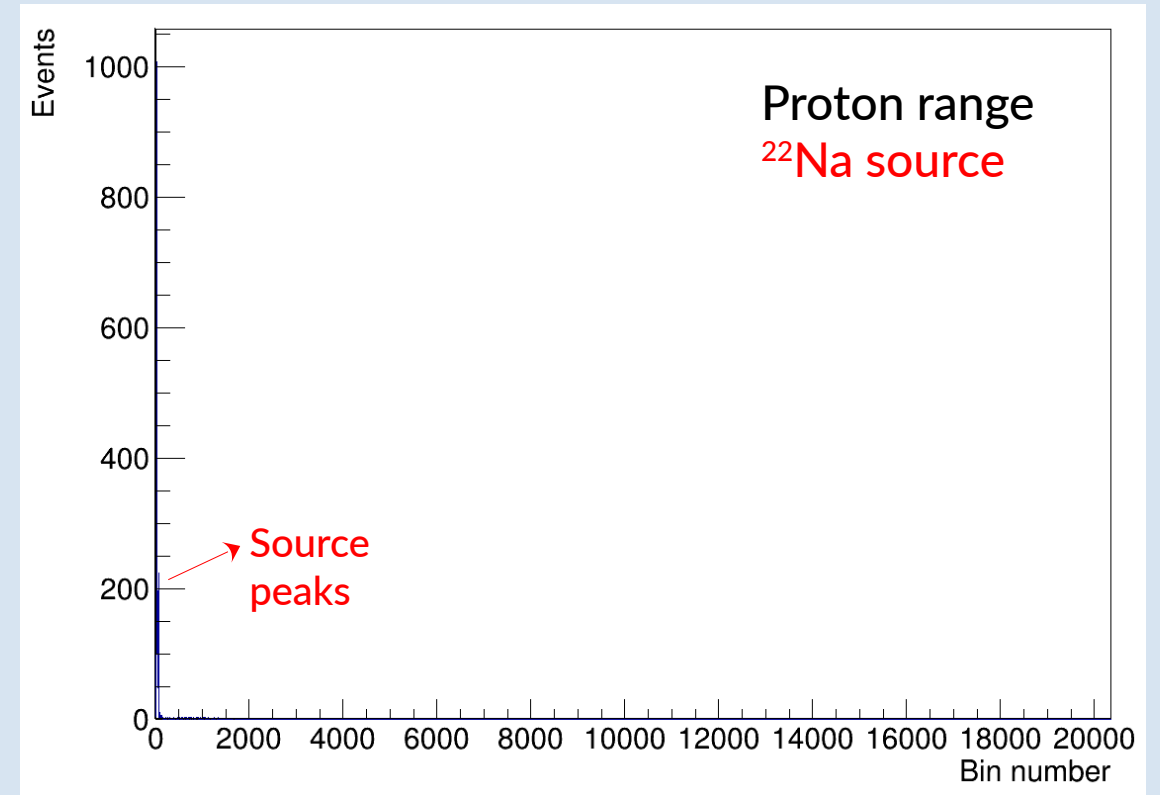
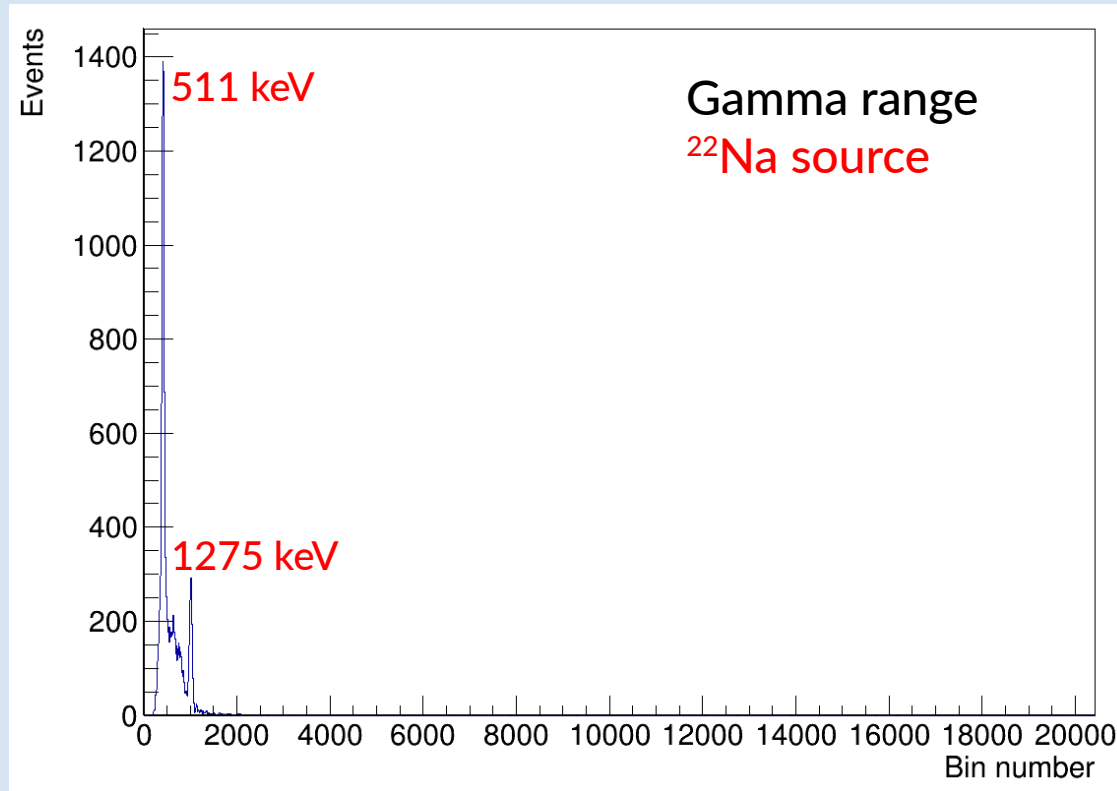
iPhos: $19^\circ < \theta < 43^\circ$

CEPA: $7^\circ < \theta < 19^\circ$

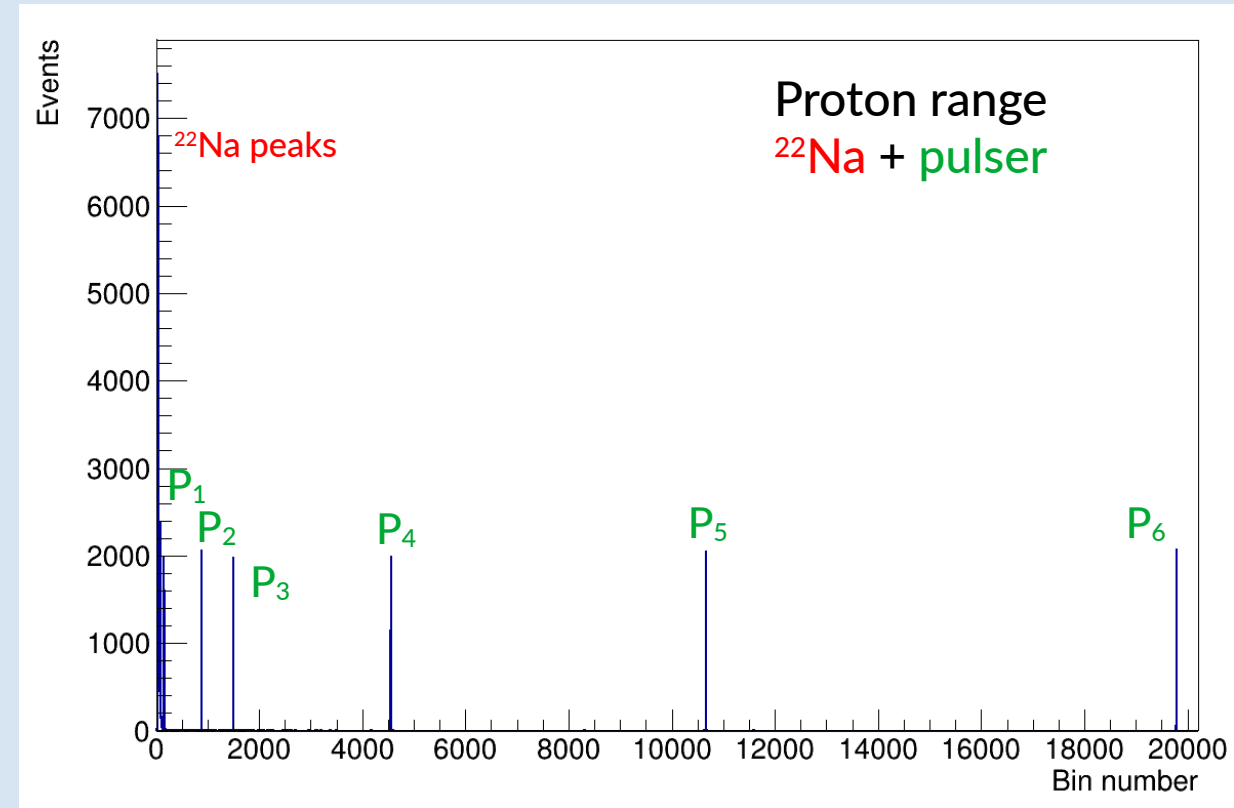
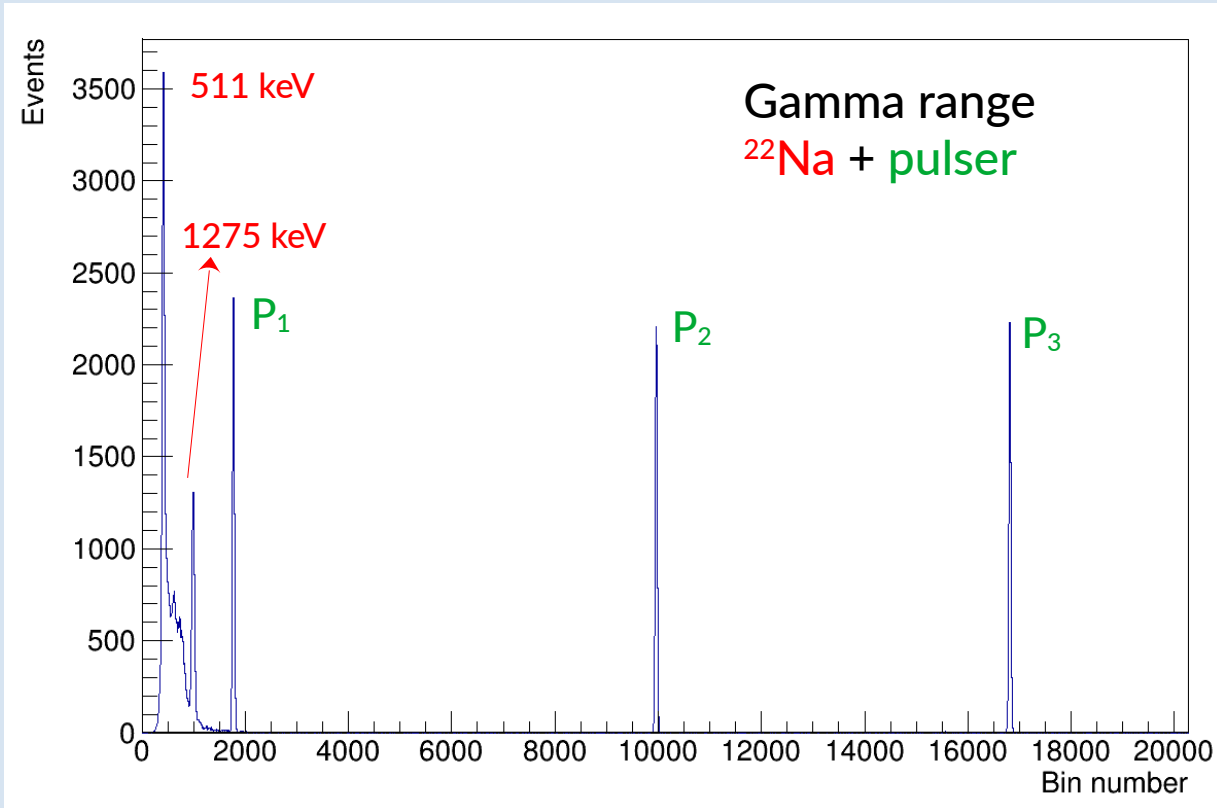


Group report:

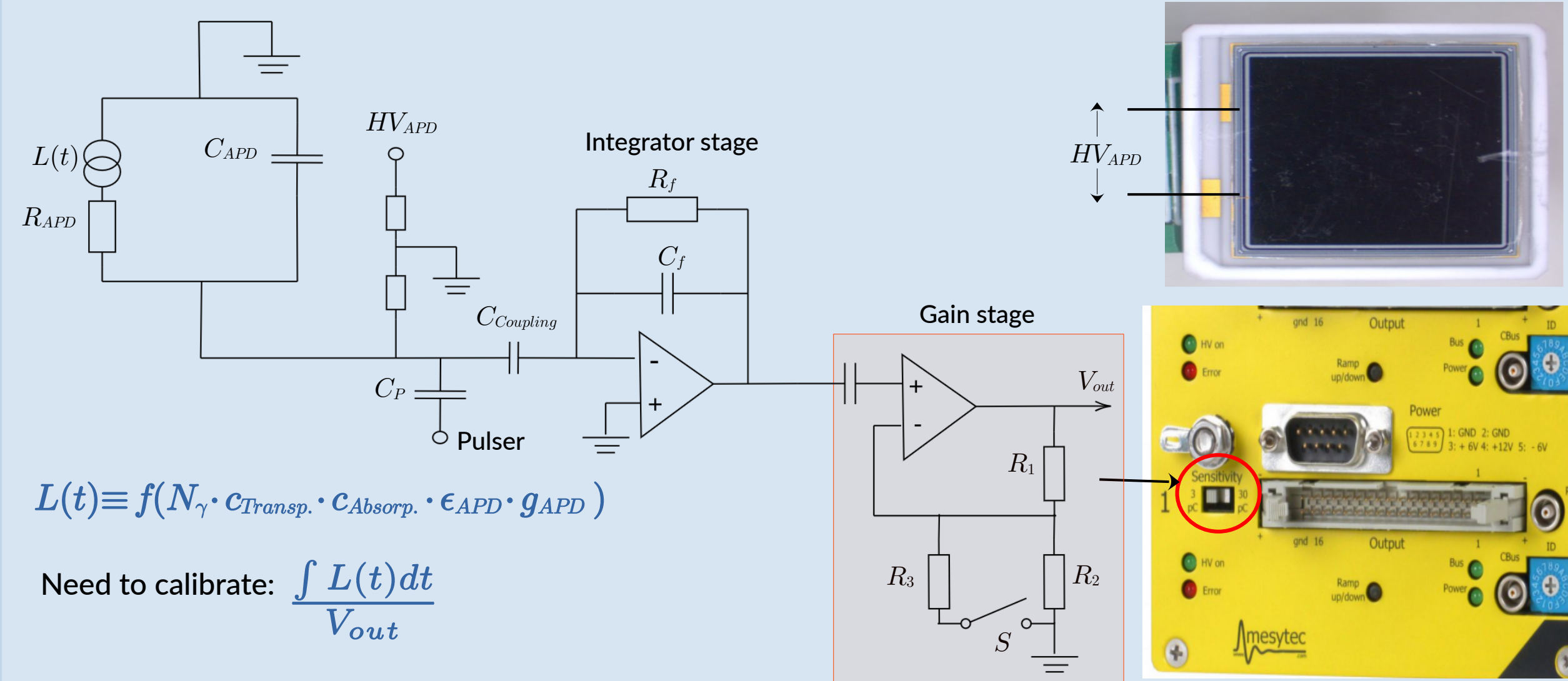
HK 33.1, Tobias Jenegger (Wed.,17:30)



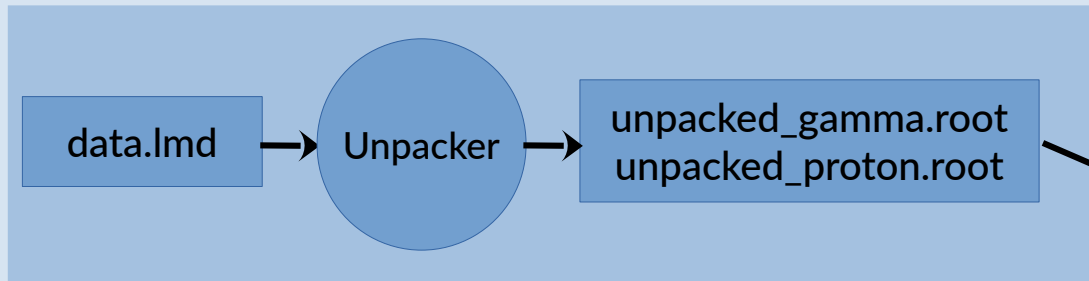
^{22}Na source hardly useful for performing calibration in the proton range !



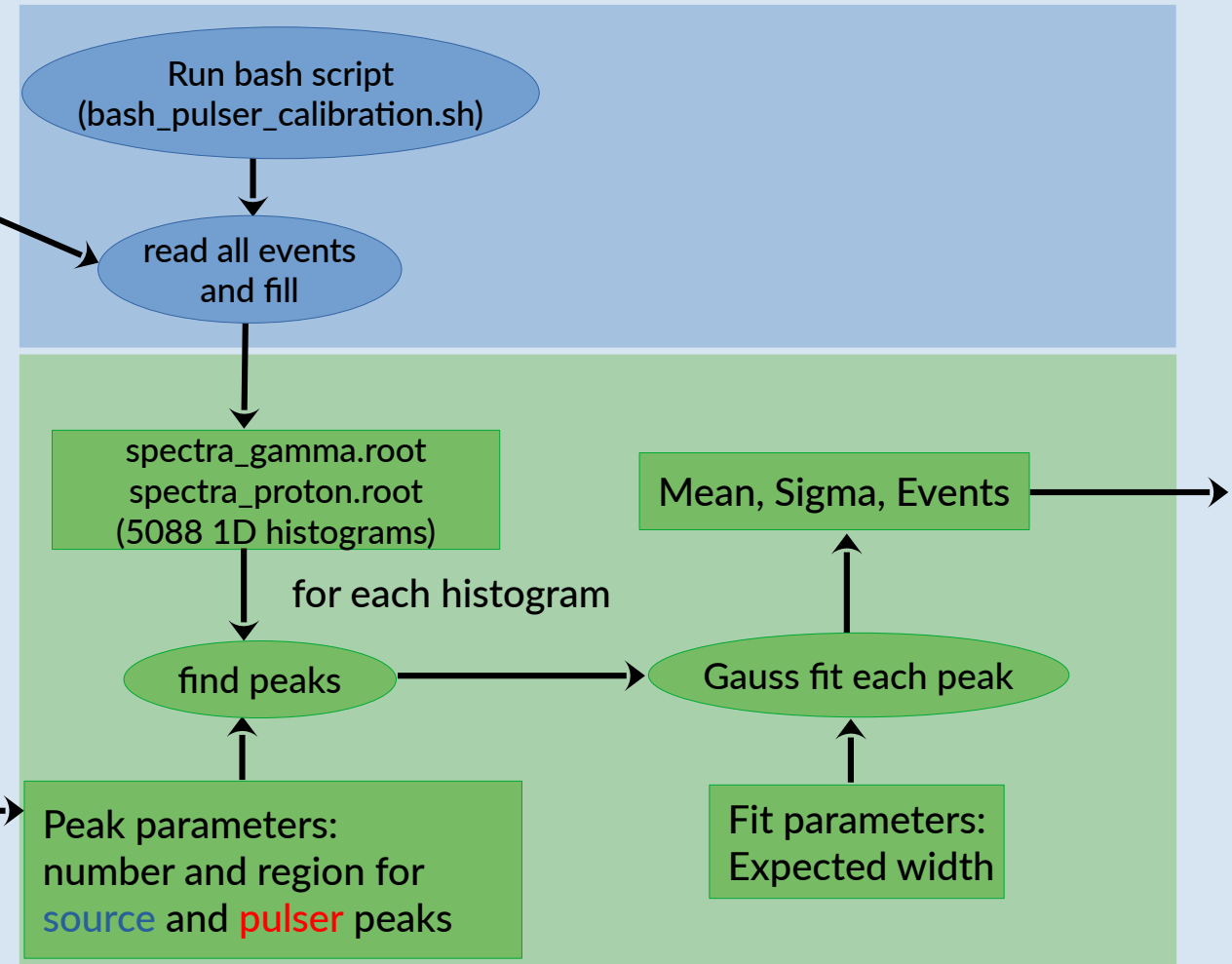
Using **pulsers** is essential for calibration in proton range



Unpacking

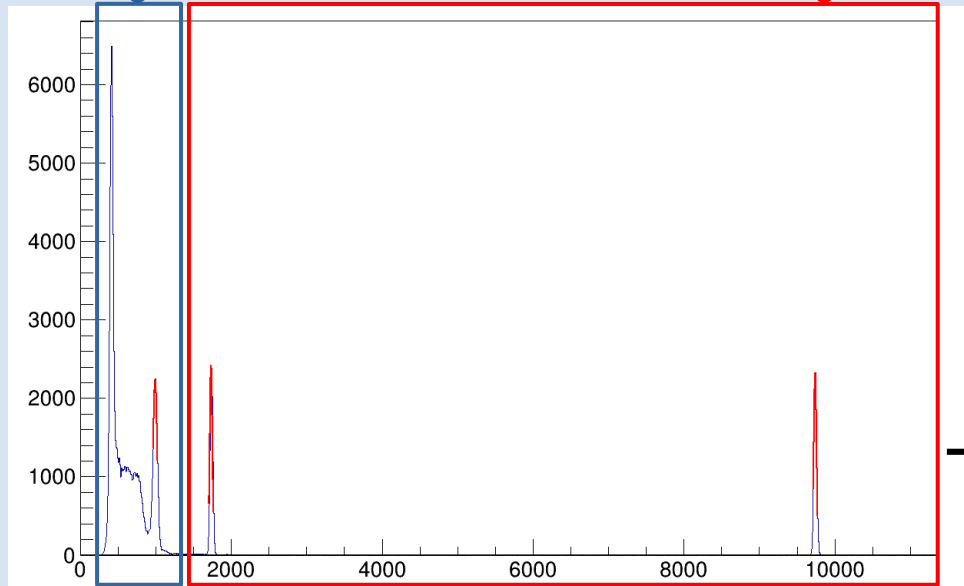


Calibration macro

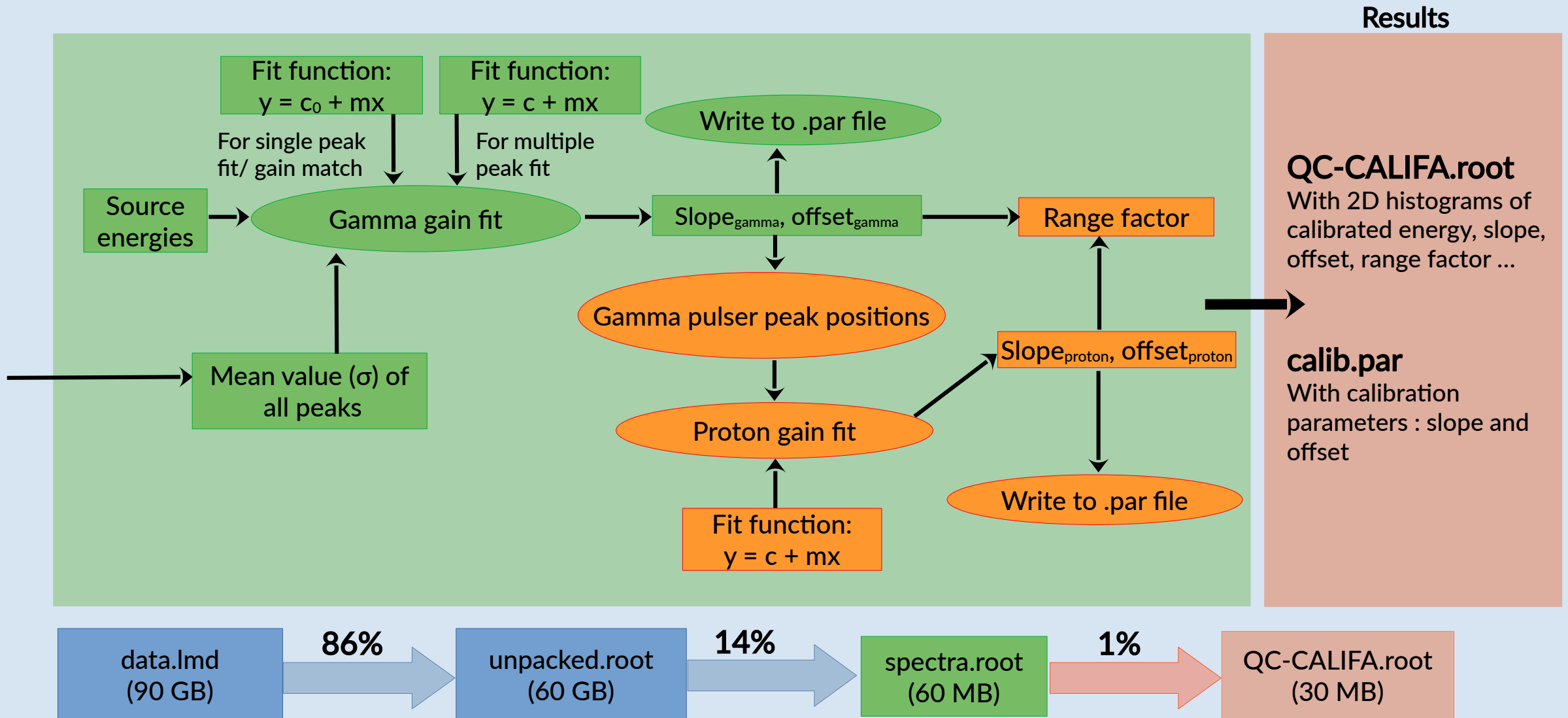


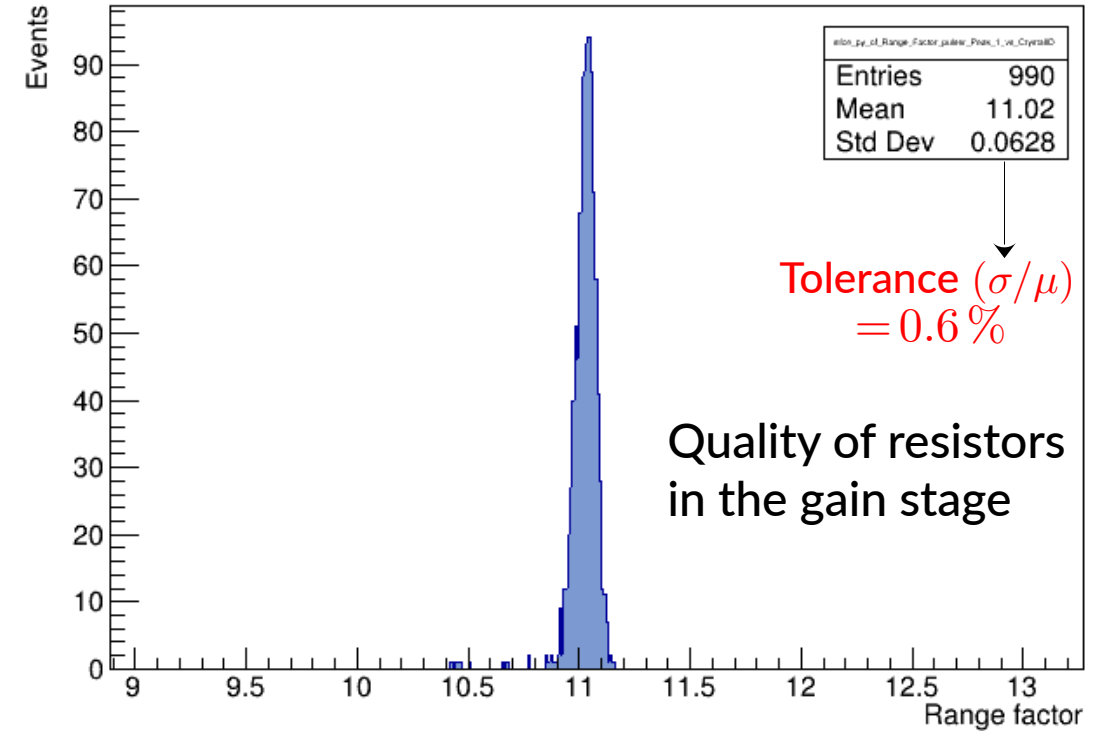
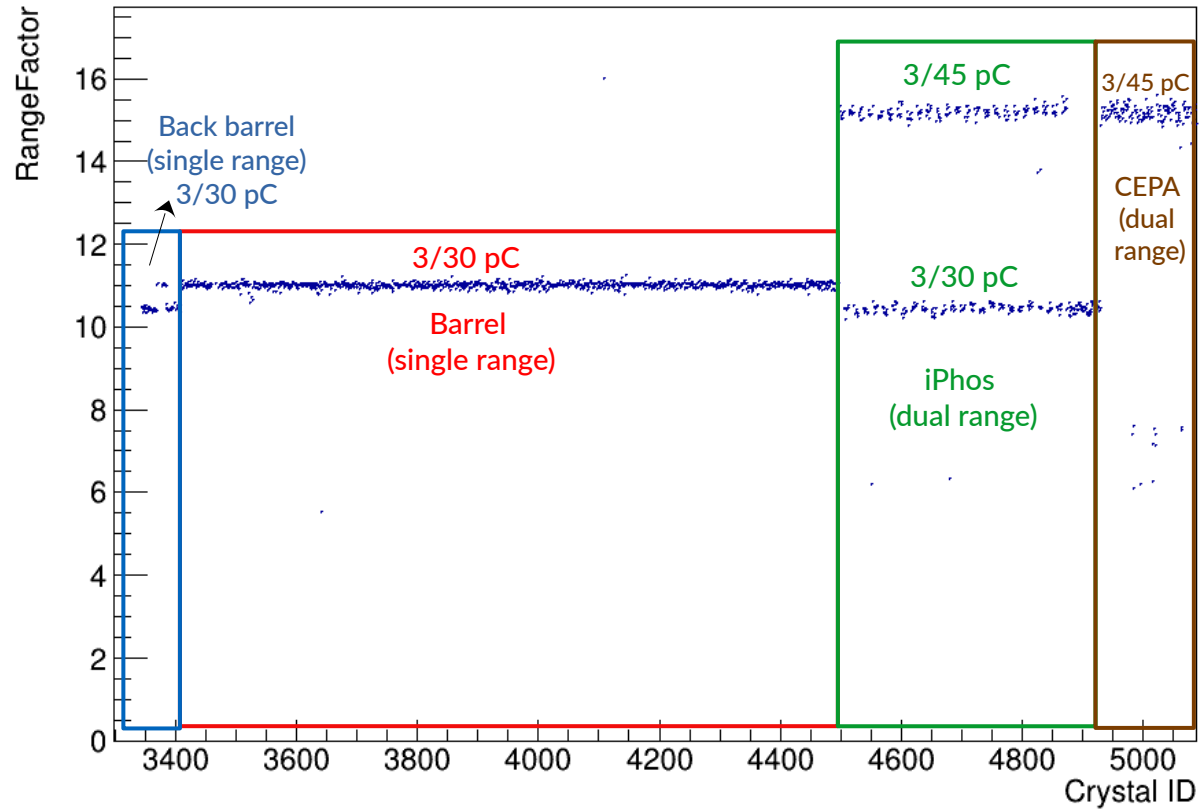
Source region

Pulser region



Calibration algorithm

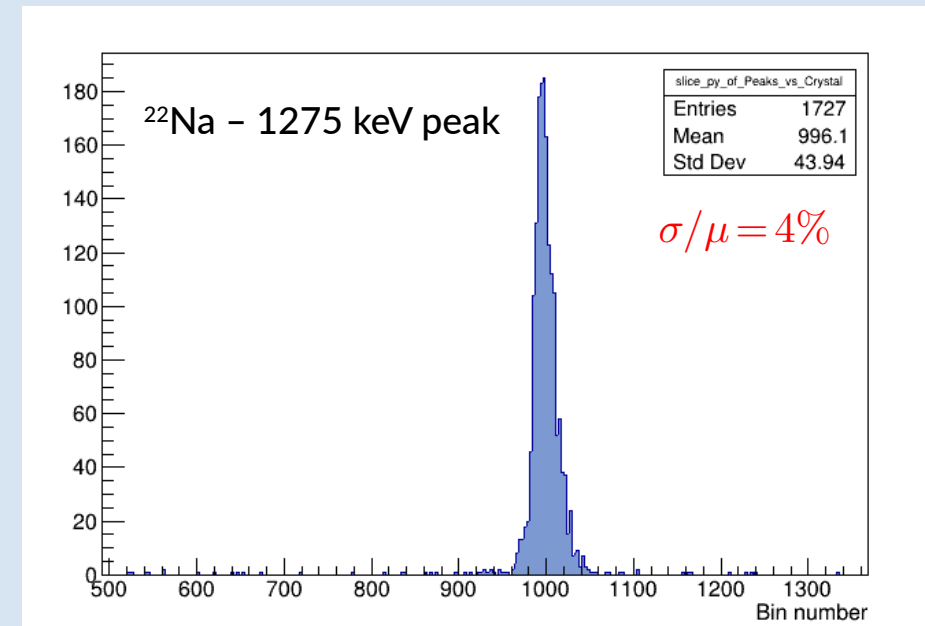
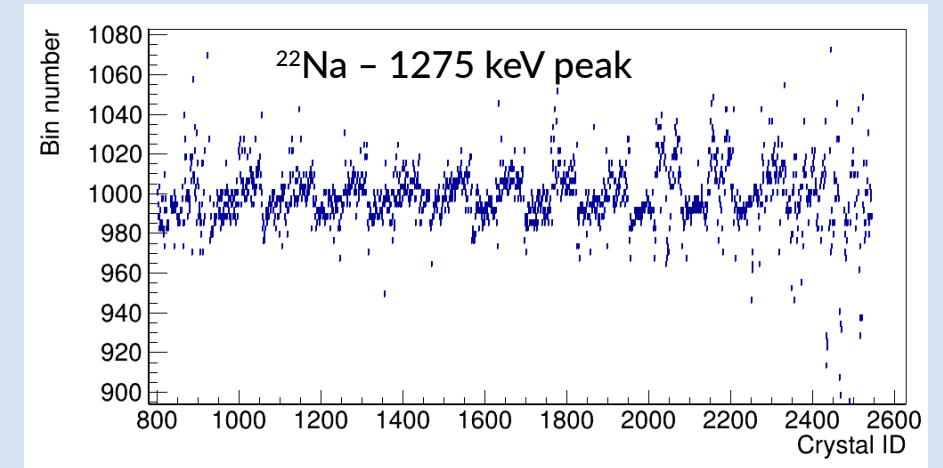
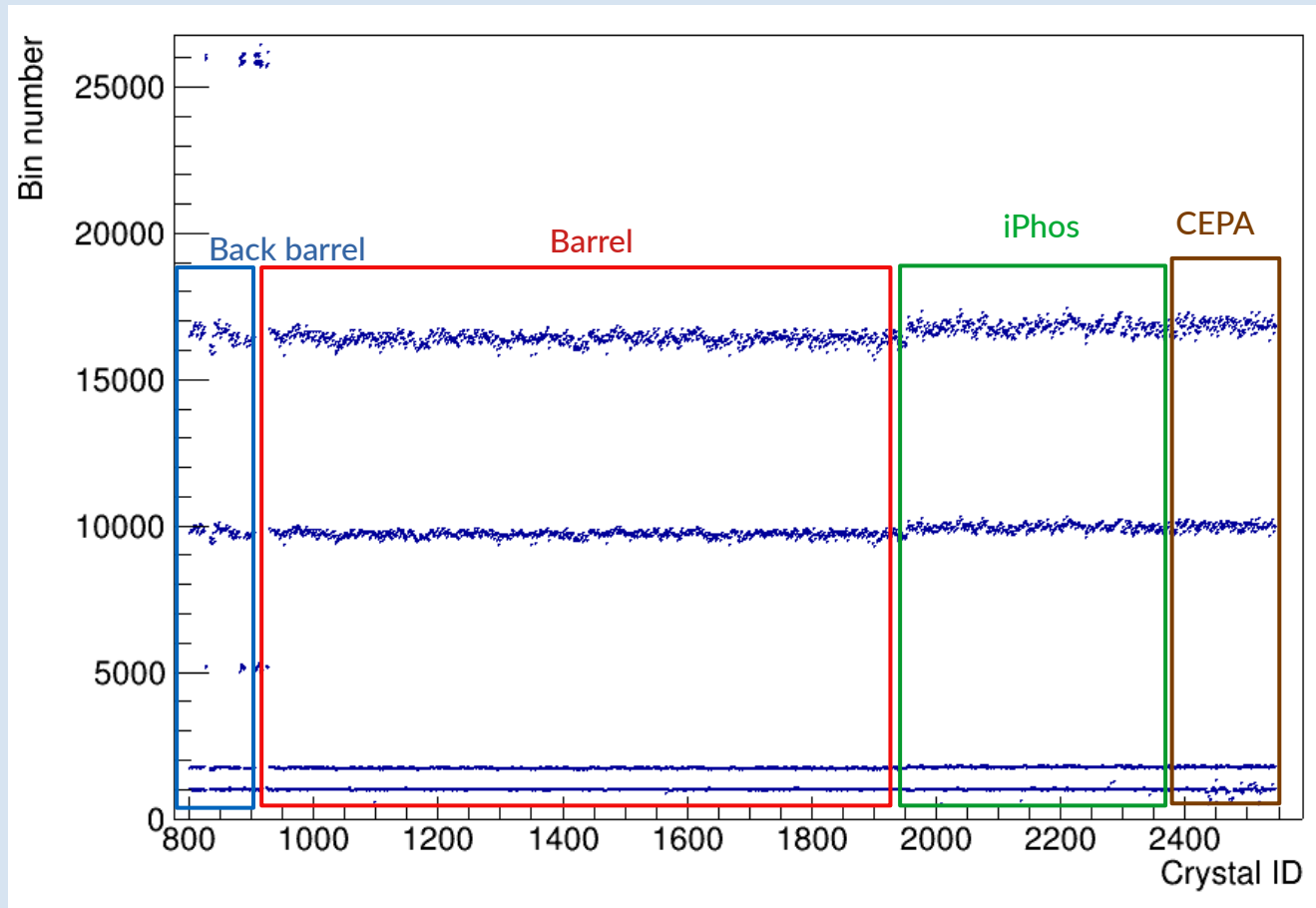


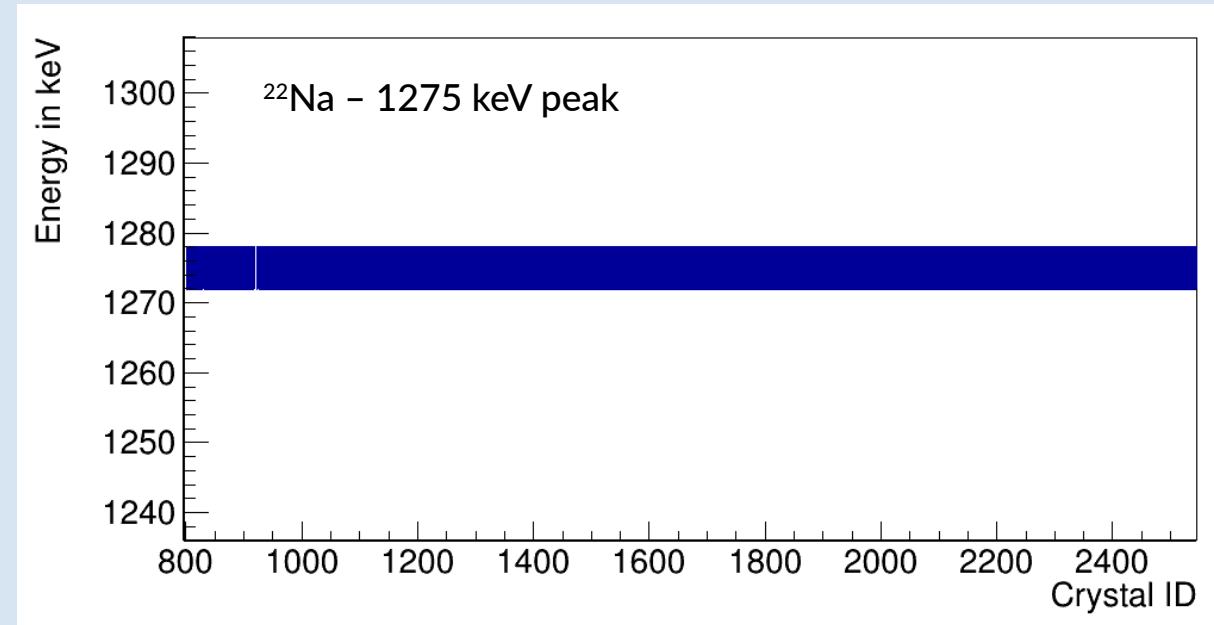
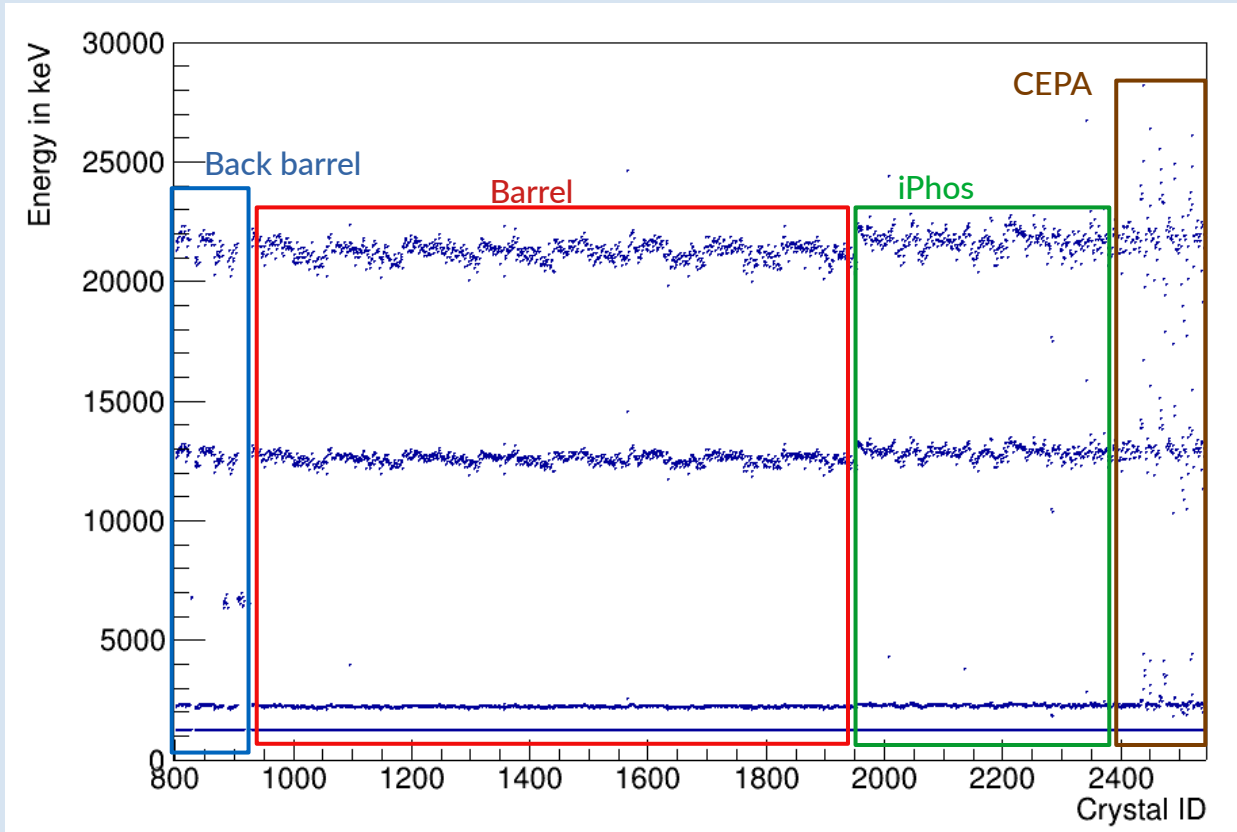


$$R = E_{\gamma(\text{uncalib.})} / E_{p(\text{uncalib.})} \approx 11$$

Serves as a translation factor between gamma range and proton range

Gamma range: uncalibrated spectra

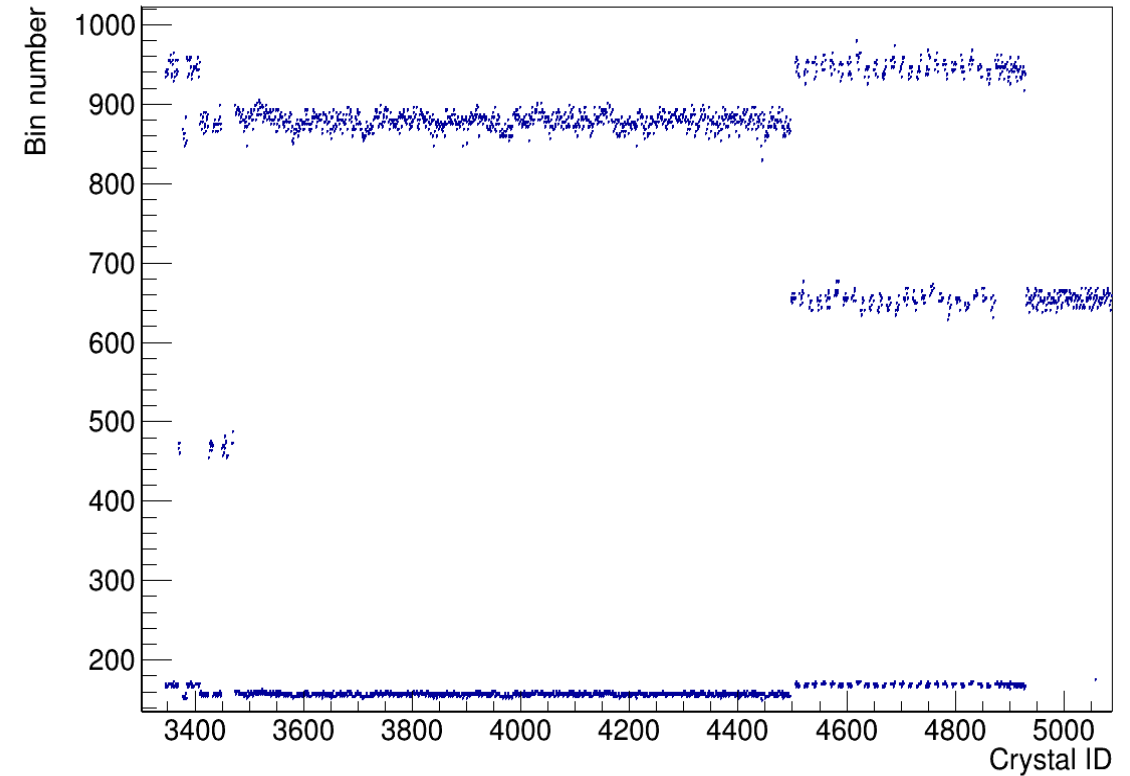
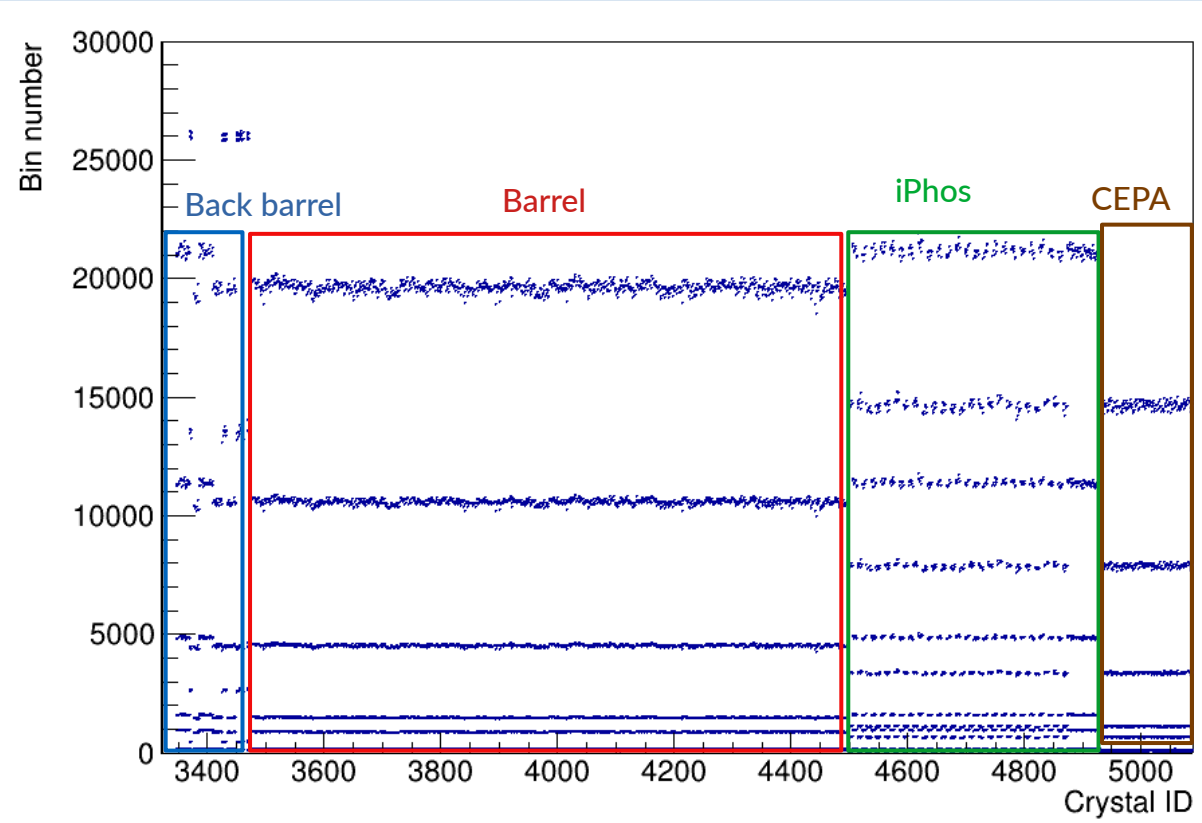




Applying a linear fit: $y = m_{\gamma}x + c_{\gamma}$

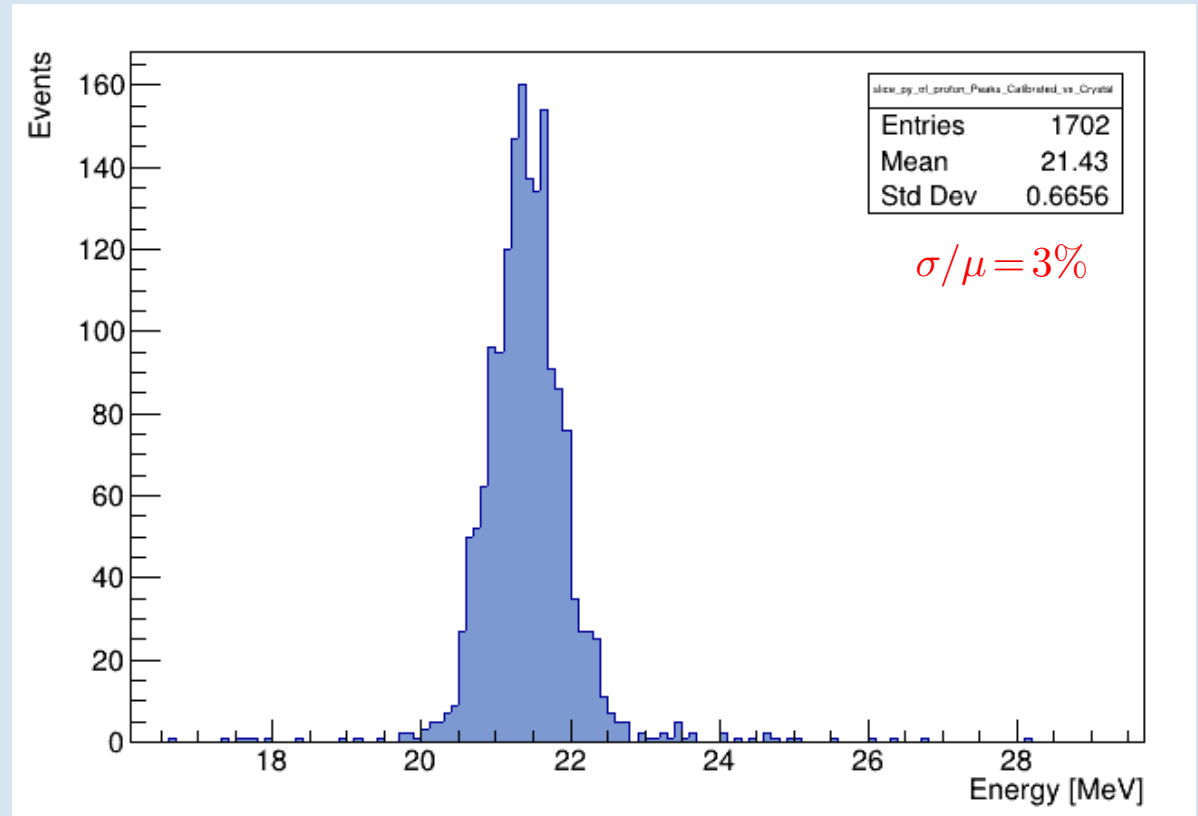
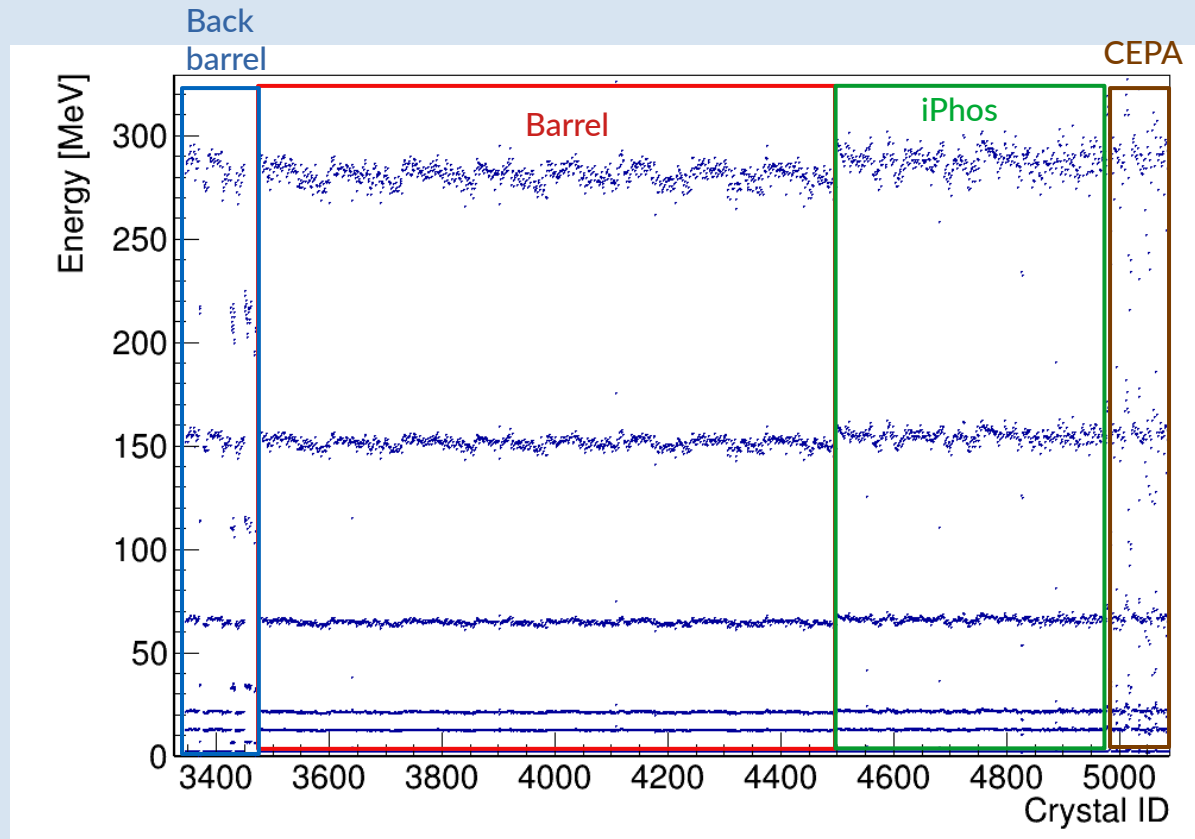
Software works !

Now we also have pulser peaks in keV



There are (channel to channel) variations in the electronics

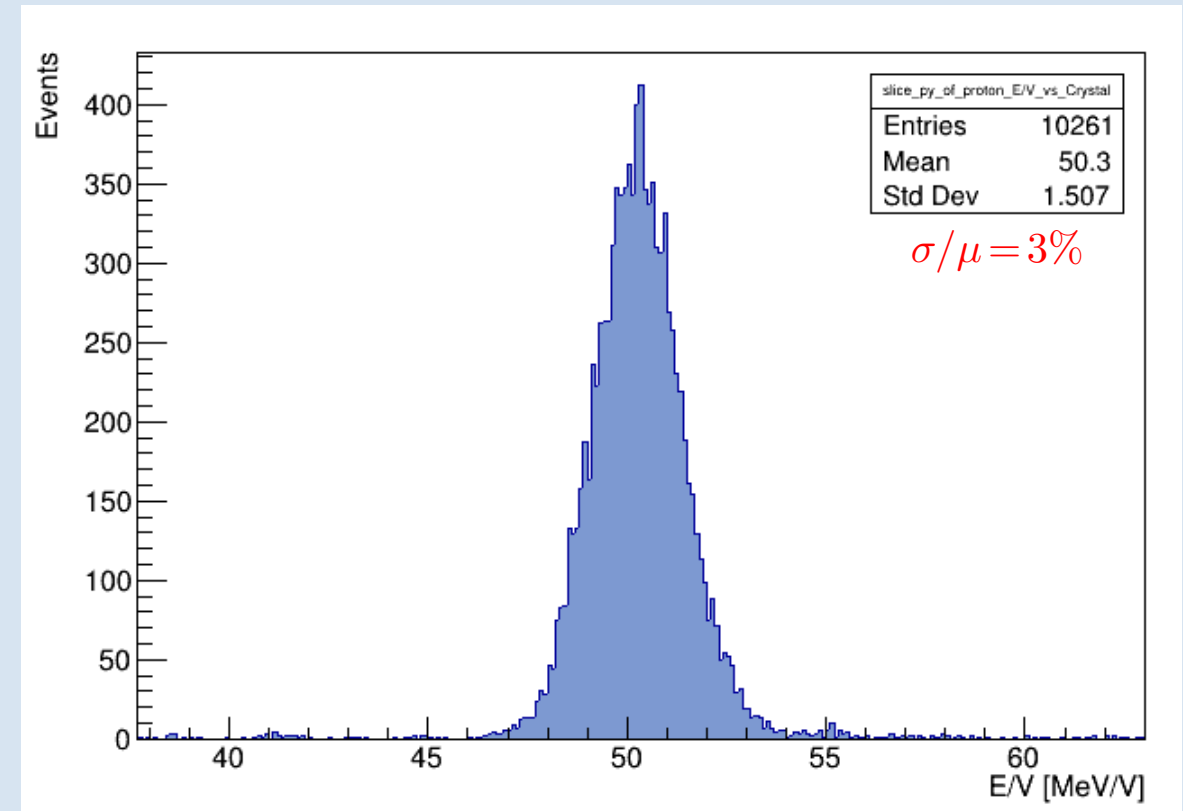
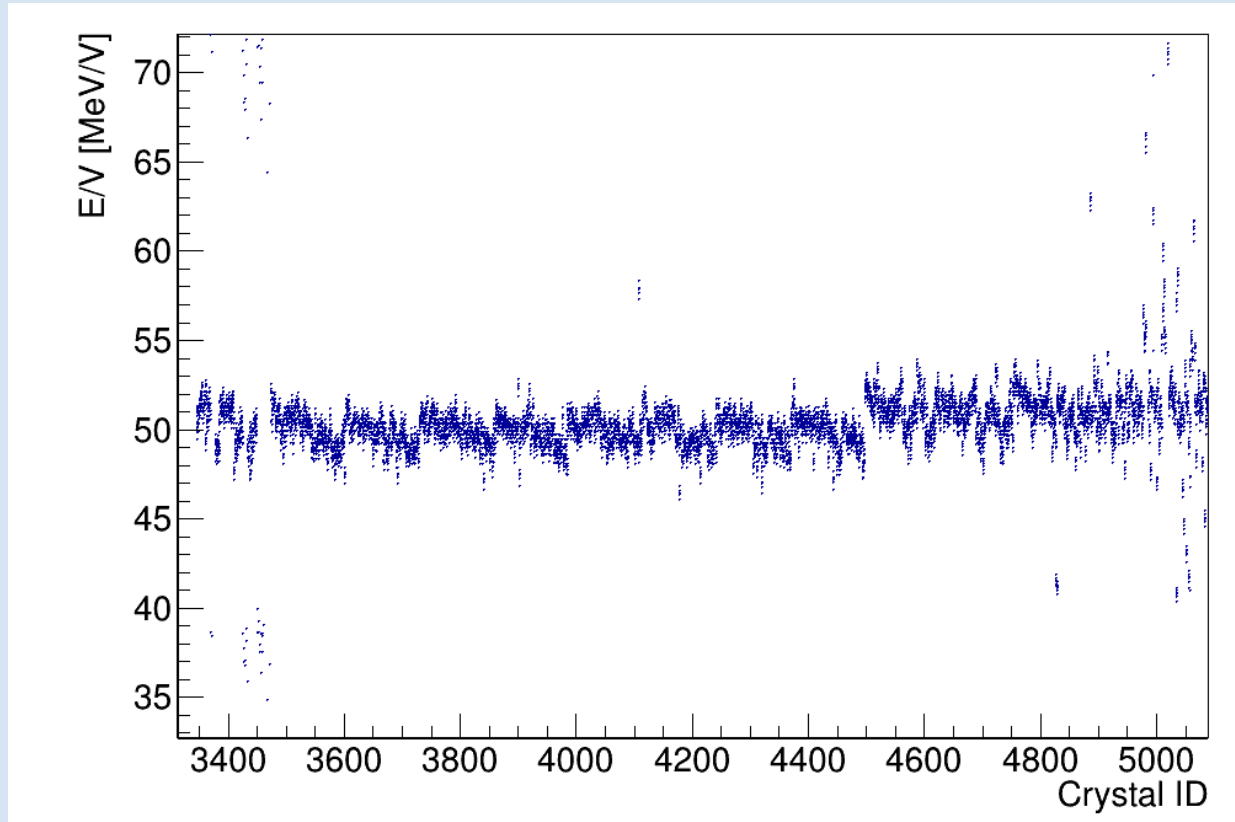
Calibration in proton range



Energies for first 3 pulsers obtained from gamma calibration

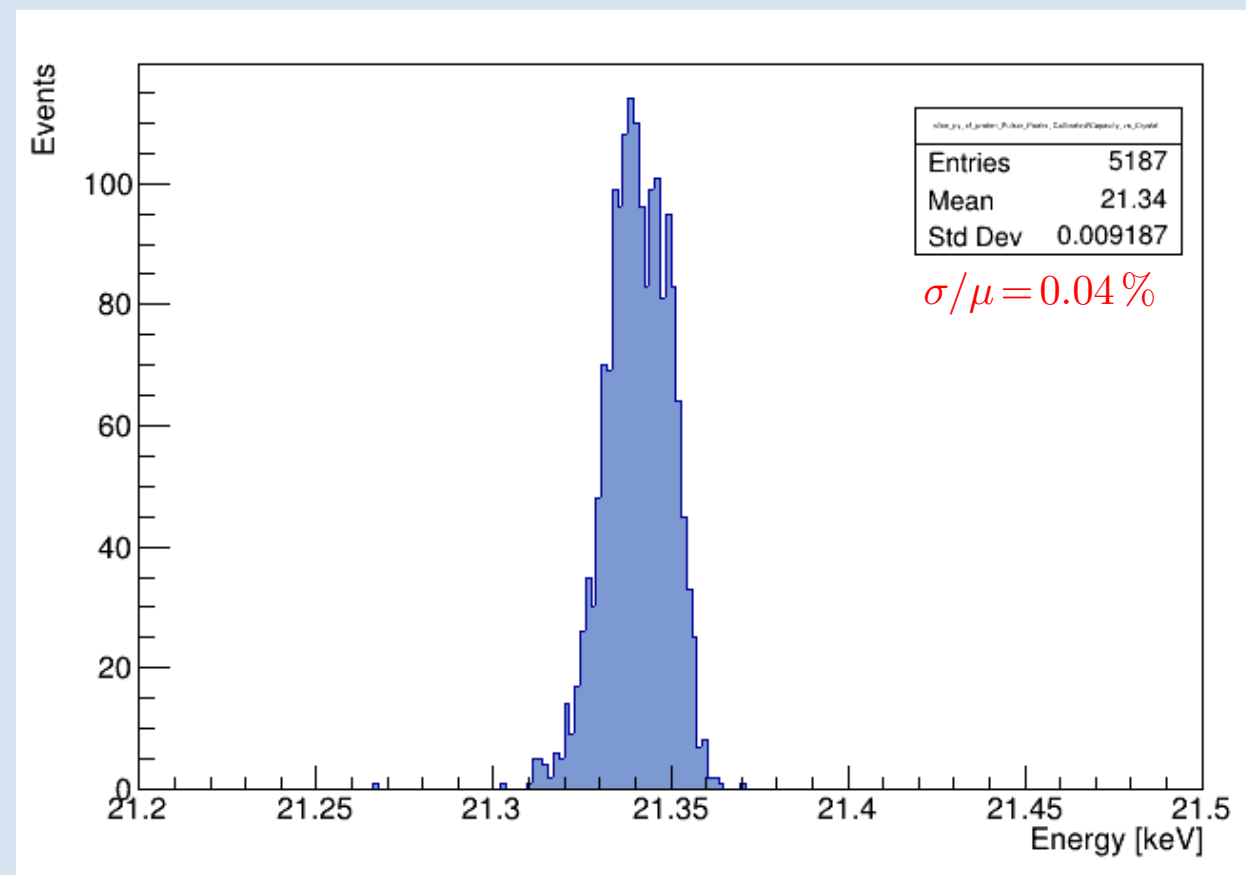
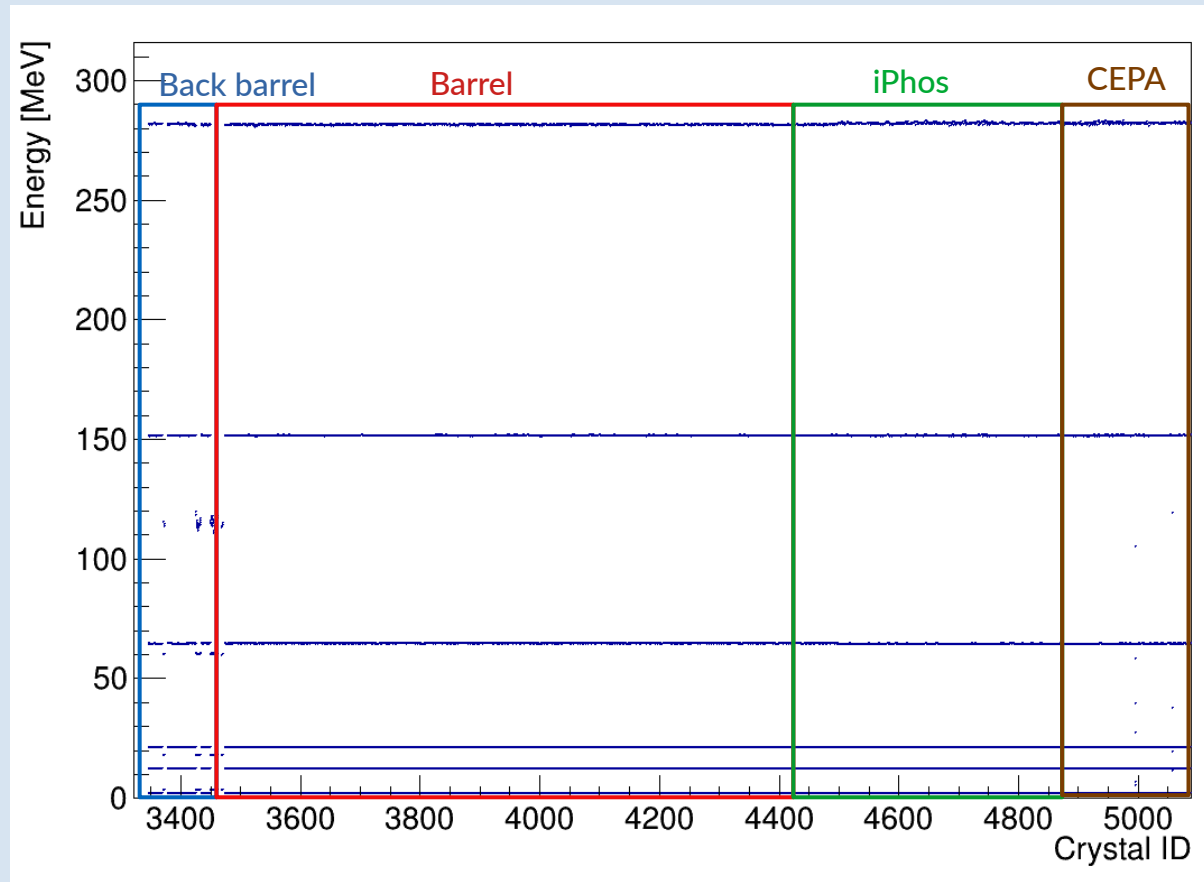
Then doing a linear fit : $y = m_p x + c_p \Rightarrow$ All pulser energies in keV

Variation in pulsers include all uncertainties (depending on $L(t)$, R and C_{pulser})

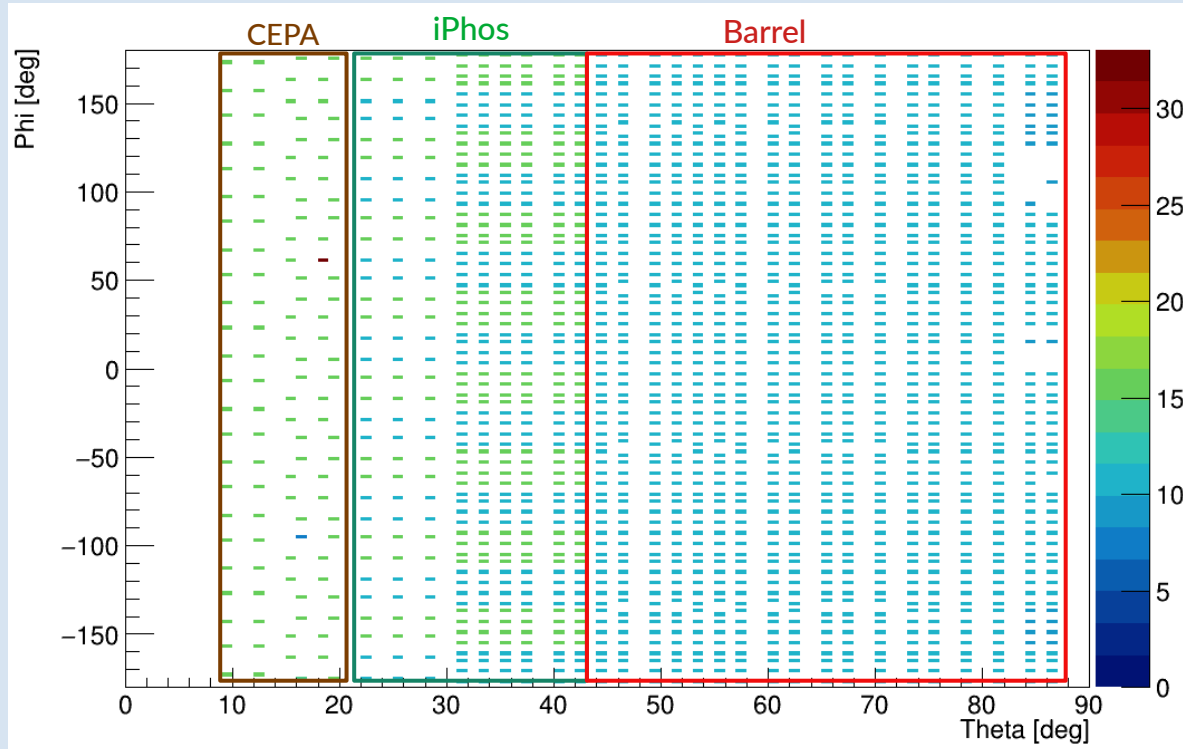


Variation in E/V for the pulsers again include all uncertainties ($L(t)$, R and C_{pulser})

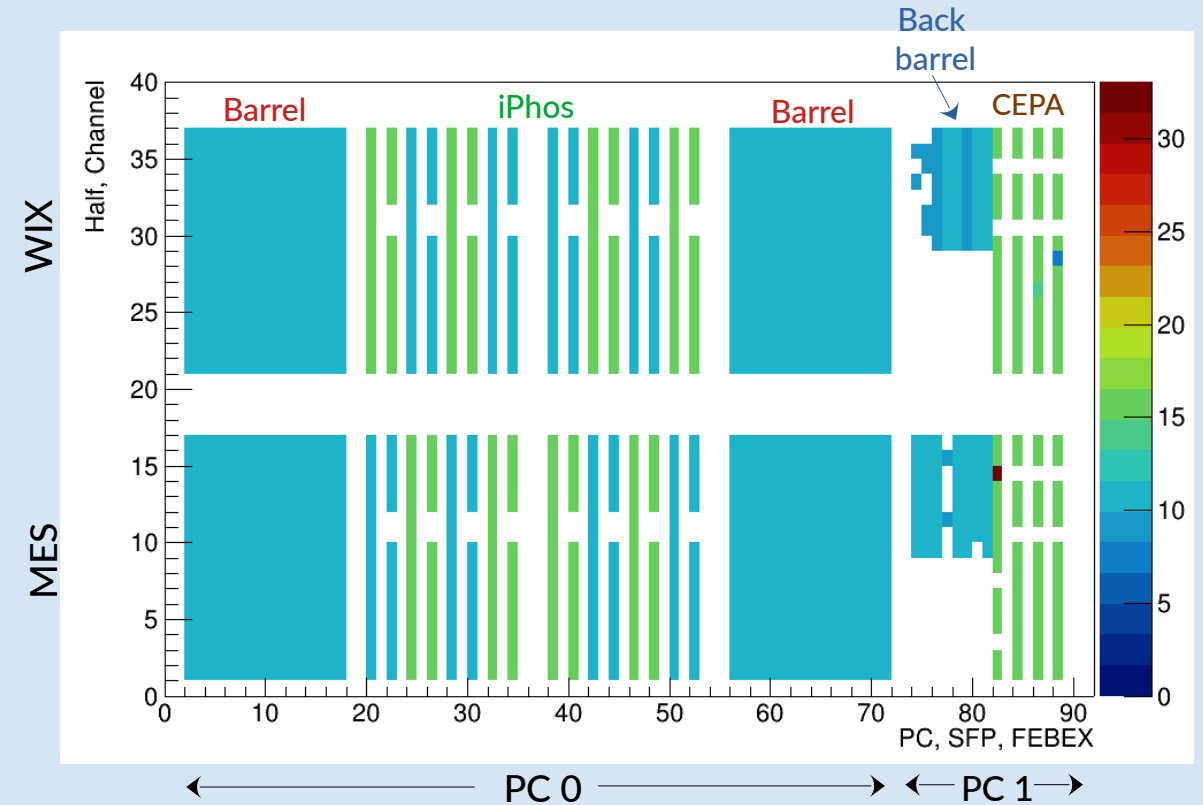
Proton range: calibrated spectra



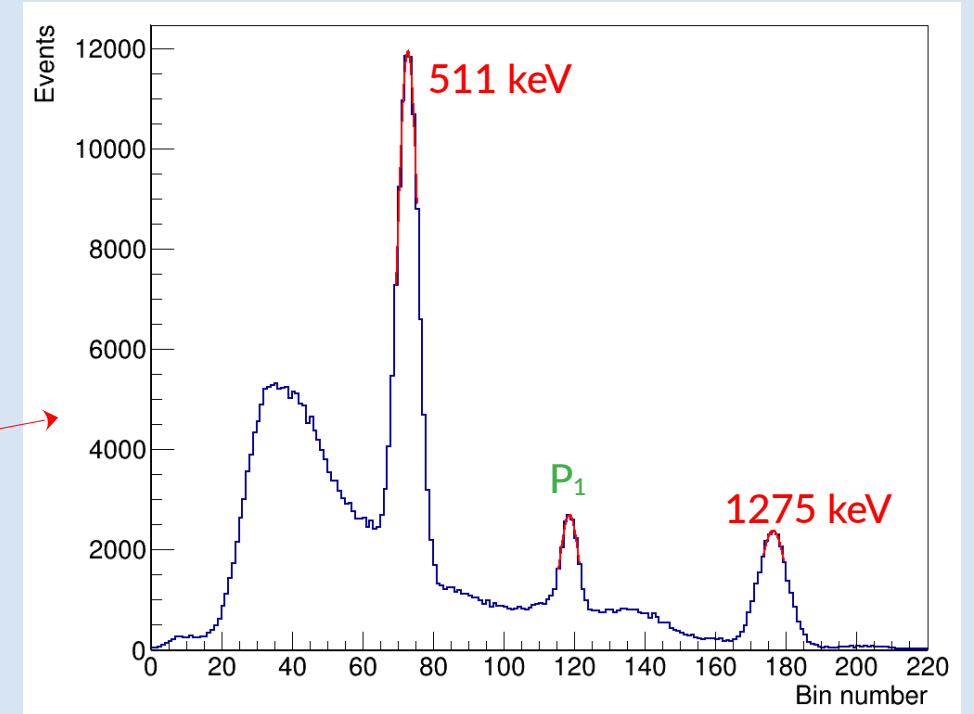
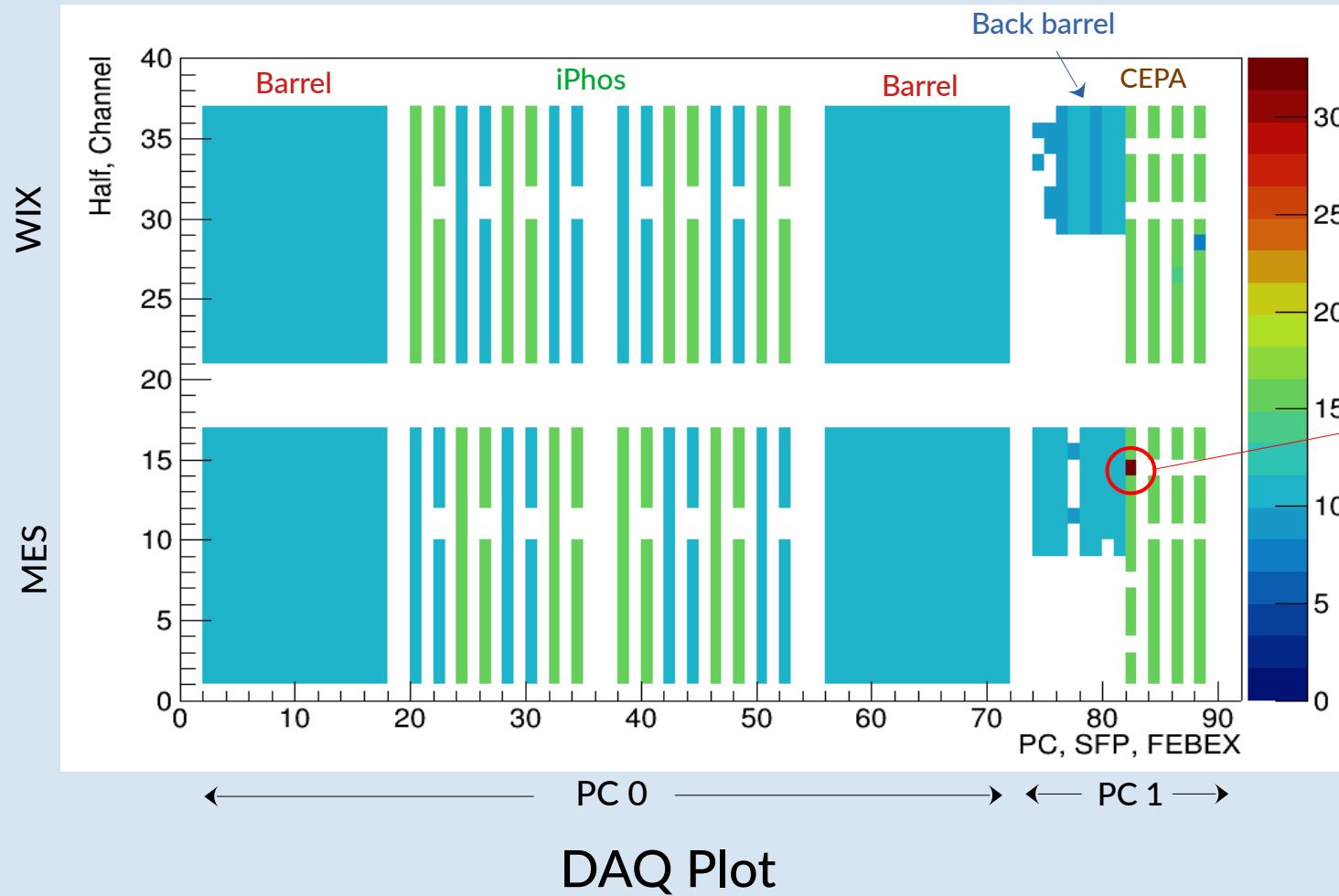
Software works !



Quality Assessment Plot



DAQ Plot (Hardware level)



Spectrum for bad channel
(incorrect gain matching)



Thank You !

CALIFA @ Technical University of Munich
Mrunmoy Jena, Tobias Jenegger, Roman Gernhäuser

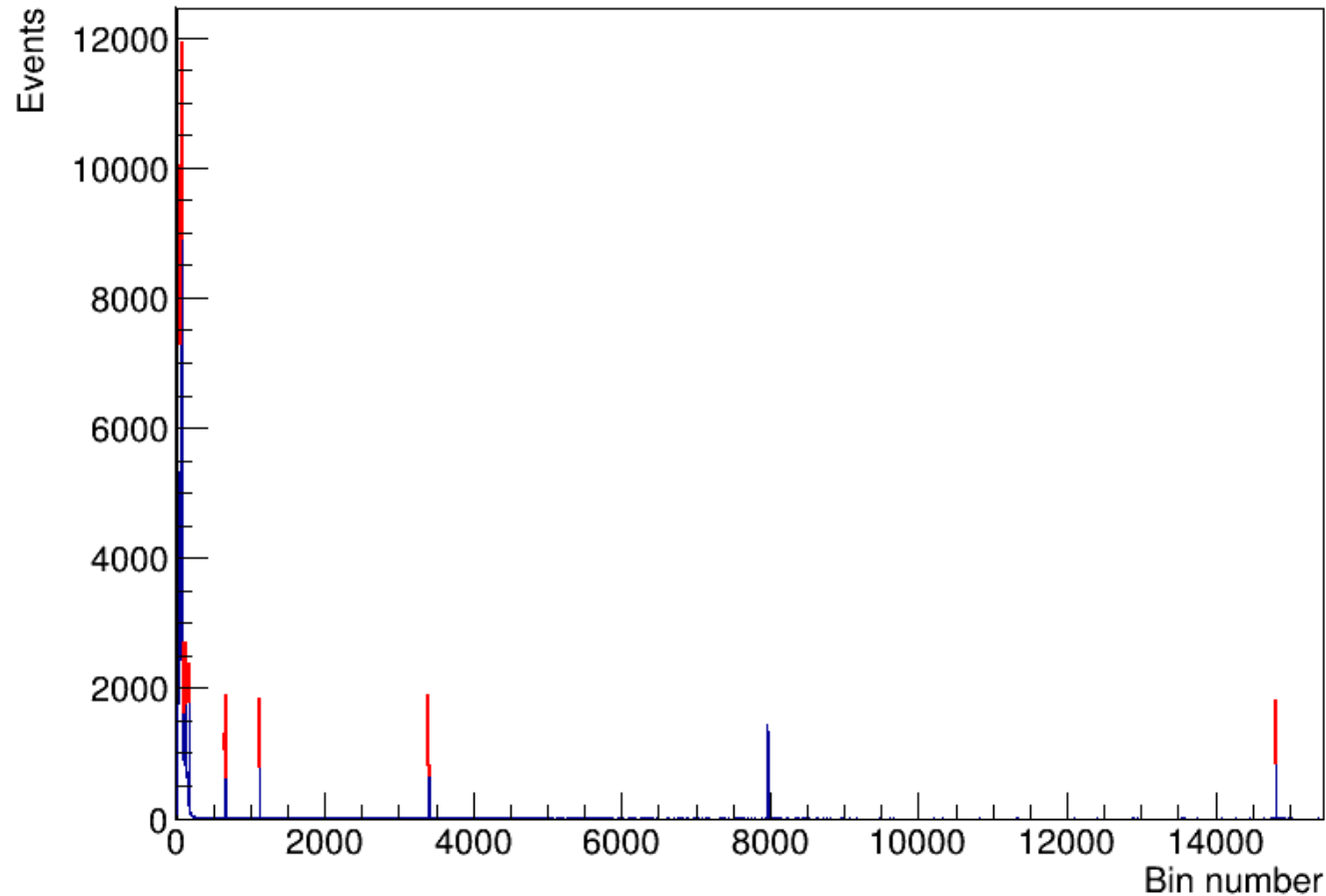


GEFÖRDERT VOM



Bundesministerium
für Bildung
und Forschung





Crystal ID: 5057, in CEPA

Crystal 5057

Source Peak 1, Bin number: 72.8852
Pulser Peak 1, Bin number: 118.622
Pulser Peak 2, Bin number: 176.335
Pulser Peak 3, Bin number: 664.026
Pulser Peak 4, Bin number: 1121.33
Pulser Peak 5, Bin number: 3402.16
Pulser Peak 6, Bin number: 14819.5
Range factor Pulser Peak 1: 15.2875
Range factor Pulser Peak 2: 57.8574
Range factor Pulser Peak 3: 25.9151
Range factor: 33.02
Pulser Offset: -2053.82
Pulser Slope: 36.794